Week 4 Interim Submission

Ethics, Safety & Risk Awareness in Healthcare Technology

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Part 1: Draft Risk Register

The following register lists eight named risks relevant to the proposed AIassisted triage support tool at Mercer General Hospital, grouped across four dimensions: AItechnical, operational, ethical, and equity. Each risk is assigned a likelihood and impact rating (L = Low, M = Medium, H = High), a proposed mitigation, and a measurable signal of success. This is a draft, five risks are fully developed for interim; the full register of 8–12 will appear in the final submission.

Risk Name	**Category**	**Likelihood**	**Impact**	**Mitigation**	**Signal of Success**
Demographic bias in training data	AItechnical	H	H	Audit training data for demographic distribution before deployment; apply reweighting or stratified sampling to correct imbalances.	Sensitivity and specificity are statistically equivalent across race, ethnicity, language, and age groups in pilot evaluation.
Distribution shift model trained on nonCaribbean data	AItechnical	H	H	Finetune the model on locally sourced Mercer ED data before deployment; retrain quarterly as patient demographics shift.	Model performance on local validation set meets or exceeds published benchmarks within one standard deviation.
NLP degradation on abbreviated freetext	AItechnical	M	H	Calibrate NLP component on a representative sample of real Mercer triage notes, including abbreviated and colloquial entries, prior to going live.	NLP classification accuracy on locally sourced notes is within 5% of accuracy on clean clinical corpus.
Alert fatigue leading to clinician disengagement	Operational	H	H	Set high confidence alert threshold (>0.85 probability); target fewer than 5 alerts per shift; conduct prelaunch calibration with Sr. Patrice Alleyne.	Alert override rate remains below 30% at four weeks postdeployment; nurses report the tool as 'helpful' on weekly usability surveys.
EHR data lag causing stale inputs	Operational	H	M	Design tool to operate on data available at triage card completion, not EHR confirmation; display data age indicator alongside each alert.	Fewer than 10% of alerts are triggered on inputs more than 15 minutes old during pilot monitoring.

Automation bias nurses deferring to AI over clinical judgement	Ethical	M	H	Train all triage nurses on AI limitations before golive; design UI to show alert as advisory, not directive; log all override decisions for audit.	Nurses correctly override the tool in at least 80% of injected test scenarios where the AI recommendation is deliberately incorrect.

Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
Lack of informed patient consent for AIassisted triage	Ethical	M	M	Display a visible notice in the triage waiting area explaining that AI tools support not replace nurse decisions; obtain ethics board clearance before pilot launch.	Ethics board approval obtained; patient The complaint rate related to AI use remains at zero during pilot.
Underperformance on paediatric and elderly patients	Equity	M	H	Stratify pilot evaluation data by age group; if performance gap exceeds 5% between adult and paediatric/elderly cohorts, restrict tool use to adult patients pending retraining.	Sensitivity gap between adult and paediatric/elderly cohorts is less than 5% in the pilot evaluation report.

Part 2: RealWorld AI Harm Case Study

Incident: Epic Sepsis Model University of Michigan Health System (2021)

What happened

The Epic Sepsis Model (ESM) is a machine learning algorithm embedded within the Epic electronic health record system and used widely across US hospitals to flag patients at risk of developing sepsis. A 2021 independent evaluation published in JAMA Internal Medicine by Wong et al. audited the ESM's performance across 38,455 patient encounters at the University of Michigan Health System. The study found that the model had a positive predictive value of just 12.8% meaning that fewer than 1 in 8 patients flagged by the algorithm actually developed sepsis. Furthermore, the model missed 33.4% of patients who did develop sepsis. The tool was in active clinical use, generating alerts that nurses and physicians were expected to act on, yet its realworld performance was substantially worse than the performance Epic had reported in its own validation studies.

Why it happened root cause analysis

- Training/deployment distribution shift. The model was trained on retrospective data from a single health system (University of Wisconsin). When deployed across institutions with different patient populations, clinical workflows, and documentation practices, performance degraded significantly. Epic's internal validation was not independently replicated before widespread deployment.
- Opaque validation reporting. Epic did not publish the model's source code, training data, or full performance statistics. Hospitals had no basis on which to evaluate whether the model was appropriate for their patient population before going live.
- Automation bias in clinical response. Despite the model's poor specificity, clinical teams in multiple sites reported feeling pressure to act on ESM alerts because they were embedded in the EHR workflow and associated with institutional sepsis protocols. The design of the alert made deferral feel like inaction.
- No signal-of-success monitoring. Hospitals lacked predefined criteria for determining when the model was underperforming in their specific context. There was no mechanism to trigger a review or suspension of the tool.

What this means for the proposed Mercer pilot

The Epic Sepsis Model case is the clearest published precedent for the risks in this proposal. Three direct design responses follow from it. First, the Mercer tool must be validated on locally sourced ED data before deployment external validation statistics cannot be assumed to transfer. Second, performance must be monitored continuously during the pilot, with predefined thresholds that trigger automatic review if sensitivity or specificity fall below acceptable levels. Third, the alert design must make it cognitively easy to override the UI should present the AI recommendation as one input among several, not as a directive requiring justification to ignore. The ESM case also reinforces the risk register finding that distribution shift and automation bias are the two highest priority risks for this pilot, both rated High likelihood and High impact.

Source

Wong, A., Otles, E., Donnelly, J. P., Krumm, A., McCullough, J., DeTroyerCooley, O., Pestrue, J., Phillips, M., Konye, J., Penoza, C., Ghassemi, M., & Singh, K. (2021). External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. JAMA Internal Medicine, 181(8), 10651070. https://doi.org/10.1001/jamainternmed.2021.2626